

CASE STUDY

Problem:

10,000 people in the US turn 65 every day and ageing leads to degeneration of the discs of the spine and chronic pain. According to WHO this low back pain is the leading cause of disability worldwide.

Goal:

To stabilise vertebral segments and promote bone fusion. This can be done by removing the diseased disc and replacing it with materials that can fuse bone above and below the disc together.

Evolution of devices used to treat the condition:

- Bone grafts without structural support
- 1980s: Introduction of titanium cages for structural support
- 1990s: PEEK cages with improved radiolucency
- Current trends: Hybrid designs and biomimetic materials

ANSWER :

Final Cage Design Structure

- ☒ **Ti Outer Shell:** For strong initial support, placed on the Apophyseal ring (cortical bone).
- ☒ **Slot-in Mg Alloy Inner Core:** Resorbable, supports adaptive modulus concept.
- ☒ **Chitosan Coating:** Slows Mg degradation to avoid premature loss of structural integrity.
- ☒ **rhBMP-2 Bone Graft Chamber:** Central chamber to stimulate osteoinduction at the point of surgery.
- rhBMP-2 coating – problem is sterilising as you need to use gamma An alternative: The use of nanoscale Ti as a coating for PEEK (just on the surface)

Component	Material	Purpose
Outer layer	3D printed Titanium	Strong support, interfaces with Apophyseal ring
Inner core	Slot-in Mg alloy , coated	Resorbable, allows load transfer over time
Bone graft chamber	Filled with rhBMP-2	Stimulates bone growth
Surface features	Concave + nanoscale	Promote osteoinduction and osteointegration
Optional features	Silver/copper coating, PVDF layer	Antimicrobial & electrical stimulation
Manufacturing	SLM/EBM printing + etching	Precision & osteoactive surface

Definitions:

- Osteointegration is the process by which a direct and stable connection forms between living bone and the surface of a load-bearing implant without intervening soft tissue.
- An artefact refers to a distortion or error in the imaging data caused by the interaction of the metal with X-rays, leading to streaks, shading, or loss of detail around the metal object.

- Radiolucent refers to a material or structure that allows X-rays or other forms of radiation to pass through it with minimal absorption, appearing dark or less dense on radiographic image.
- Bone on growth refers to the formation of new bone directly on the surface of an interbody cage, while bone ingrowth involves new bone growing into the porous structure of the cage, enhancing mechanical interlocking and stability.
- Osteoconduction is the process by which stem cells migrate along a framework, allowing for osteoblast attachment and ultimately bone formation.

To treat this, spinal fusion is performed: the degenerated disc is removed and replaced with an **interbody fusion cage**, which encourages bone growth between adjacent vertebrae. However, current designs often:

- **Shield the bone from stress** (causing bone weakening),
- Are **too stiff**, preventing bone from taking load (violating Wolff's Law),
- And/or **block imaging**, making fusion progress difficult to assess.

What is Wolff's Law? - Bone becomes stronger when it's used, and weaker when it's not.

- When your bones are **regularly loaded with weight or stress** (e.g., walking, running, lifting), they respond by **getting stronger**.
- If bones don't experience stress (e.g., you're lying in bed for weeks, or using a very stiff implant), they become **weaker and less dense**.

Mechanical Performance & Safety

- The cage must initially support **spinal load** during typical movements like bending or lifting.
- The **Apophyseal ring**, composed of strong cortical bone, should interface with the **load-bearing parts** of the cage.
- **Cancellous bone** in the vertebral body center is softer and should be paired with **resorbable, less stiff materials**.

Best approach:

- Use **Titanium (Ti)** for the outer structure due to its strength and biocompatibility.
- Avoid Ti in solid form due to **stress shielding** and **imaging artefacts**.
- Use **3D printed lattice structures** to reduce stiffness and mimic cancellous bone.

1. Patient Safety & Adaptive Modulus

To prevent early implant failure and encourage healing:

- Use **two complementary materials**:
 - **Outer layer**: Permanent, strong (e.g. **3D printed titanium**) placed on the **Apophyseal ring** (strong cortical bone).
 - **Inner layer**: **Resorbable**, softer (e.g. **magnesium alloy**, aligned with the cancellous bone core).

☑ **Adaptive modulus**: As **Mg degrades**, load transfers gradually to the new bone, encouraging it to bear weight in line with **Wolff's Law**.

Safety measure: Coat Mg with **chitosan** to **slow degradation**, preventing premature mechanical failure.

Key Principles for Patient Safety:

1. Mechanical Compatibility:

- Use **strong outer layer (Ti)** to bear high initial loads — placed on the **Apophyseal ring** (strong cortical bone).
- Use **soft, resorbable inner core (Mg)** aligned with the **cancellous bone** inside the vertebra.

2. Adaptive Modulus:

- Mg **degrades gradually**, transferring stress from the implant to the new bone.
- Coated with **chitosan** to prevent premature mechanical loss.
- Surgeon can personalise the insert: **fast / medium / slow** resorption options.

3. Patient-Specific Design:

- **Slot-in inserts** allow customisation during surgery depending on patient's bone quality.
- Ensures a tailored balance of **support vs stimulation**.

Bone Graft Chamber (in centre of cage):

- Can be filled during surgery with **rhBMP-2**, which:
 1. **Stimulates bone growth**.
 2. **Provides scaffold** for osteoblasts.
 3. **Promotes fusion** between vertebra

Structural Design for Safety:

- **Porosity and Lattice Architecture:**
 - Reduce stress shielding.
 - Improve nutrient flow and integration.
- **Antimicrobial Coatings:**
 - **Silver:** Prevents infection, safe for bone regrowth.
 - **Copper:** May enhance angiogenesis, but used cautiously.

2. Driving Bone Formation

Porosity and Shape

- **Porous lattice structures** improve nutrient flow, reduce stiffness, and allow bone ingrowth.
- **Concave pores** mimic **osteoclast resorption pits**, promoting osteoblast activity — a **biomimetic design** proven to induce bone even in non-bony areas.

Surface Features

- **Submicron/nanoscale roughness** increases surface area, protein adsorption, and cell attachment.
- Achievable via **acid etching on titanium** post-printing.

Biological Additions

- Central **bone graft chamber** to load **rhBMP-2** at surgery, encouraging bone growth (osteoinductive).
- Optional addition of **piezoelectric materials** (e.g. **PVDF**) that create microcurrents during movement, enhancing osteogenesis.

Material	Pros	Cons
Titanium (Ti)	Strong, good osseointegration, easy to coat and etch	Radiopaque, stress shielding if solid
PEEK	Radiolucent, flexible	Hydrophobic, poor cell adhesion, hard to shape
Mg Alloy	Biodegradable, supports bone formation, less radiodense	May degrade too fast if uncoated
PLA	Clinically tested	Acid by-products, poor cell response, radiodense
CFR-PEEK	Strong, lightweight	Hard to 3D print, risk of fibre migration

4. Radiolucency and Imaging

- **Imaging is essential** to assess bone fusion.
- **Metals (Ti)** cause artefacts; **PEEK** and **Mg** are more radiolucent.
- Best approach: an **adaptive radiolucent design** where inner resorbable parts (e.g. Mg) **degrade over time**, making fusion more visible.

5. Antimicrobial Coatings

- **Silver**: Safe in small doses, prevents biofilm, used in trauma/dental implants.
- **Copper**: May stimulate angiogenesis by mimicking hypoxia but less studied.
- Optional use depending on infection risk.

Manufacturing Method

- **Selective Laser Melting (SLM)** or **Electron Beam Melting (EBM)** are ideal for 3D printing **Ti cages**:
 - Enables precise, patient-specific lattice geometry.
 - Allows post-processing (etching) for nanoscale roughness.

After printing:

- Stress relief → heat treatment.
- Surface enhancement → acid/alkaline etching.
- Sterilisation → gamma radiation, autoclave, or ethylene oxide.

Patient Safety & Load Transfer

Q: Explain how spinal cage design ensures patient safety during and after implantation.

→ (Based on Slide 2 + "Apophseal ring" slide)

What to mention:

- Apophseal ring = strong cortical bone, so hard permanent materials like **Titanium** are used on the outside.
 - Cancellous bone in the centre = soft → softer, **resorbable inner layer** like **Mg alloy** used inside.
 - Load-bearing shifts over time with **adaptive modulus** → supports spine early, transfers stress to bone later.
 - Use **chitosan coating** to slow Mg degradation so the cage doesn't weaken before bone is strong enough.
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2. How to Promote Bone Formation

Q: What features in the cage design help maximise bone formation?

→ (Slides on Bone Formation, Topology Optimisation, Design Considerations)

Key points:

- **Concave pores:** mimic osteoclast pits → osteoblasts love them! - Biomimetic cues that encourage osteoblast attachment.
- **Porosity:** allows nutrients, waste removal, and vascularisation. CELL MOVEMENT
- **SUBMICRON & Nanoscale surface features:** improve protein adsorption, attract cells, boost osteoinduction.
- **rhBMP-2 graft:** added in central chamber → triggers bone growth.
- **HA coatings:** improve bone-on growth, especially on Ti or PEEK. (this wasn't on JOHN's slide but yes on james)

Preclinical Evidence:

- Bone formed **without chemical cues**, based on **concave shape alone**.
- **Nano features alone** led to osteoinduction in 6 weeks.

The cage promotes bone formation through concave pores, which mimic osteoclast pits and encourage osteoblast attachment — a biomimetic design. Porosity allows for the movement of cells, nutrients, and waste, which is essential during healing. Submicron and nanoscale surface features increase protein adsorption and support osteoinduction. The central chamber contains rhBMP-2, a powerful growth factor that triggers bone formation. HA coatings further improve bone-on growth, especially on Ti or PEEK. To ensure patient safety, a titanium outer shell provides strong support, while a coated magnesium inner core gradually degrades, allowing stress transfer to bone. This adaptive modulus approach supports healing while reducing the risk of stress shielding.

3. Adaptive Modulus Design

Q: What is an adaptive modulus and how does it improve long-term spinal cage performance?

→ (Slides 1, 2, and final "State of the Art Design" slide)

Answer structure:

- **Definition:** Adaptive modulus = cage stiffness changes over time.
- **Why needed?:**
 - Bone grows slowly; cage supports early.

- As bone forms, cage should "get out of the way".
 - **How achieved:**
 - Strong outer Ti shell bears initial load.
 - Inner Mg insert **gradually degrades**, transferring load to new bone (aligned with **Wolff's Law**).
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4. Materials Comparison & Justification

Q: Compare different materials used for spinal cages and justify your final design.

→ (Slides on Materials Choice + Radiolucency)

What to say:

- **Titanium:** Strong, good integration, causes CT artefacts. Best for outer shell.
 - **PEEK:** Radiolucent, but poor bone bonding and cell adhesion.
 - **Mg alloy:** Biodegradable, supports bone growth, less radiodense → better imaging. Must be coated (e.g. chitosan) to prevent fast degradation.
 - **PLA:** Poor resorption control, acidic byproducts, less preferred.
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5. Wolff's Law and Stress Shielding

Q: What is stress shielding and how does your design minimise it?

→ (Your uploaded mechanics slides + design logic)

Answer key points:

- **Wolff's Law:** Bone strengthens under load.
 - If cage is too stiff → **stress shielding** → bone doesn't grow → weak fusion.
 - **Porous lattice structure** reduces stiffness.
 - **Mg core degrades** → bone gradually takes more stress = optimal stimulation.
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6. Antimicrobial Features

Q: Why and how are antimicrobial coatings used in spinal cage design?

→ (Notes & Coatings slide)

Include:

- **Silver:** Prevents infection (biofilm), safe at correct doses.
 - **Copper:** May boost **angiogenesis**, helps vascularised bone healing.
 - Avoid high concentrations that inhibit osteoblasts.
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7. Radiolucency for Post-Op Assessment

Q: Why is radiolucency important and how is it achieved in your design?

→ (Radiolucency slide)

Points:

- Radiolucency allows surgeons to **see bone healing on X-ray/CT**.
- Metals (like Ti) block imaging = artefacts.
- Use **resorbable Mg inside** → degrades over time = clearer imaging later.
- Can be an **adaptive radiolucent design** — radiopaque at first, radiolucent later.

Titanium offers excellent mechanical strength and is ideal for load-bearing outer structures, but it is radiopaque and can cause stress shielding. PEEK is radiolucent and lightweight but has poor bone integration due to its hydrophobic, bioinert surface. Magnesium alloys are bioactive and support bone formation while being partially radiolucent, but they degrade quickly and may require coatings like chitosan to ensure safety. An adaptive radiolucent design — using a Ti outer and a Mg inner core — provides strong initial support while allowing improved imaging and gradual load transfer. This combination ensures both mechanical performance and clinical safety.

Q: Discuss how material and structural design choices in spinal cages ensure patient safety and effective spinal fusion.

Answer:

Patient safety in spinal cage design relies on achieving a balance between **mechanical stability**, **biological compatibility**, and **controlled degradation**. This ensures that the implant provides immediate support post-surgery while gradually transferring the load to the newly forming bone, aligning with **Wolff's Law** — the principle that bone remodels and strengthens in response to mechanical stress.

The cage is designed with a **biomimetic, two-layer structure** to match the anatomy and mechanics of the vertebra. The **outer layer**, which interfaces with the **Apophyseal ring** (a region of strong cortical bone), is made from **3D-printed titanium**. This material offers high strength and can be post-processed to develop **submicron surface features** that support osseointegration. Its stiffness is appropriate for cortical bone, reducing risk of implant failure and ensuring mechanical compatibility.

The **inner layer** consists of a **slot-in resorbable insert** made from a **magnesium alloy**, which is aligned with the softer, **cancellous bone** at the centre of the vertebra. Magnesium gradually degrades over time, reducing the implant's stiffness — an effect referred to as **adaptive modulus**. This gradual reduction allows the mechanical load to be transferred to the healing bone, encouraging bone formation and preventing **stress shielding**. To prevent the magnesium from degrading too quickly, it is coated with **chitosan**, which slows degradation and synchronises it with the rate of bone growth.

To support patient-specific safety, the design includes **personalised degradation options**, allowing the surgeon to select **fast, medium, or slow resorbing inserts** during surgery based on the patient's bone condition. This ensures that the cage can be tailored to the biological response of each patient.

A **central chamber** is included in the cage design for **intraoperative delivery of rhBMP-2**, a potent bone growth factor. This chamber promotes osteoinduction, enhances the fusion process, and ensures early biological stability.

Additional safety features include a **porous lattice structure**, which lowers the stiffness of the titanium, reduces stress shielding, and improves nutrient diffusion. The implant may also be coated with **antimicrobial agents** such as **silver** or **copper**. Silver is already clinically used and is safe for bone regrowth, while copper may enhance angiogenesis by mimicking hypoxic conditions, although it is less widely validated.

Together, these features ensure that the spinal cage provides early structural support, adapts to the healing process, and promotes effective, safe spinal fusion tailored to individual patient needs.

How Design Features Promote Bone Formation

◆ **1. Porous Structure**

- Allows **nutrients, blood, and bone cells** to move through.
- Supports **vascularisation** and **waste removal**.

- Critical when **materials like Mg degrade** (need flow to prevent build-up).
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◆ 2. Concave Pores (Biomimetic Cues)

- Mimic **osteoclast resorption pits** (where bone naturally rebuilds).
 - **Osteoblasts prefer curved areas** → even in low-load zones.
 - Proven to **stimulate bone growth** by shape alone.
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◆ 3. Submicron/Nanoscale Surface Features

- Increase **surface area** → better **protein adsorption** and **cell attachment**.
 - Can **trigger bone formation** even **without chemicals** (shown in muscle pouch studies).
 - Works well with materials like **Ti** via post-processing (acid etch).
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◆ 4. Surface Coatings

- **HA (Hydroxyapatite)** → promotes bone growth on Ti or PEEK.
 - **Nanoscale Ti on PEEK** → combines good bioactivity (Ti) with good imaging (PEEK).
 - **Permanent surface charge** → attracts natural growth factors like BMPs.
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◆ 5. Biological Additions

- **rhBMP-2** added in **central graft chamber**:
 - Stimulates bone-forming cells (osteoblasts).
 - Speeds up fusion between vertebrae.
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◆ 6. Piezoelectric Materials (e.g. PVDF)

- Generate **electric signals** when patient moves.
 - Signals attract **osteoblasts** and mimic natural healing responses.
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◆ 7. Antimicrobial Coatings

- **Silver**: Prevents infection, doesn't harm bone growth.
- **Copper**: May improve **angiogenesis** (blood vessel formation), needs more evidence.